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CHENNAI

Programme	:	**B.Tech. (BVD)**	Semester	:	**Fall 2026-27**
Course	:	**AI and Machine Learning for IC**	Code	:	**BEVD402L**
			Class Nbr	:	**CH2026270102565**
Faculty	:	**Dr. G. Lakshmi Priya**	Slot	:	**D1+TD1**
Time	:	**Date of Submission: 20/09/2026**	Max. Marks	:	**30**

DIGITAL ASSIGNMENT II – SEPTEMBER 2026

Deep Learning-Based Lithography Hotspot Detection on ICCAD-12 Benchmark Datasets

Duration: 15 Days | Maximum Marks: 30 | Group Size: 3 Students

1. Common Research Problem

Develop, evaluate, and experimentally validate an improved machine-learning/deep-learning methodology for lithography hotspot detection using the ICCAD-12 benchmark datasets, with emphasis on improving detection reliability, representation quality, computational efficiency, generalization, explainability, robustness, or decision-making.

2. Common Research Context

All 19 groups will work on the same overall research problem and ICCAD-12 dataset. The benchmark contains five datasets/benchmarks and is highly imbalanced because hotspot patterns are much fewer than non-hotspot patterns. Therefore, balanced accuracy and confusion-matrix-based measures must be emphasized rather than relying on ordinary accuracy alone.

The reference work supplied for this assignment investigates VGG16-based transfer learning with classical classifiers and a lightweight CNN trained from scratch. The lightweight CNN reported strong balanced accuracy and substantially lower inference time. Students must use this work as a baseline/reference point and must not simply reproduce its methodology.

3. Common Requirements for All Groups

Component	Common Requirement
Application	Lithography hotspot detection
Dataset	ICCAD-12 benchmark dataset; five benchmarks
Classification	Binary classification: Hotspot (HS) vs Non-Hotspot (NHS)
Baseline	Common lightweight CNN baseline / faculty-provided baseline
Primary metric	Balanced Accuracy
Additional metrics	Precision, Recall/Sensitivity, Specificity, F1-score
Visual analysis	Confusion matrix and relevant plots
Experimental design	Train/validation/test with clearly documented protocol
Research comparison	Baseline versus proposed methodology
Research independence	No dependency on code, features, results, or outputs of another group

4. Independence of the 19 Groups

Each group must complete its project independently. No group is expected to obtain intermediate results, extracted features, trained models, code, or datasets prepared by another group. The only common elements are the problem definition, ICCAD-12 dataset, evaluation protocol, and common baseline. Every group must independently formulate its hypothesis, implement its variation, conduct experiments, analyze results, and draw conclusions.

5. Common Dataset and Evaluation Requirement

All student groups shall use the **five ICCAD-12 benchmark datasets** provided for this assignment.

Each benchmark shall be treated separately for the primary experiments. Students must:

- Analyze the hotspot/non-hotspot class distribution of all five benchmarks.
- Apply the assigned ML/DL methodology to all five benchmarks.
- Use the common baseline model for comparison.
- Evaluate the baseline and proposed method using identical experimental conditions.

- Report results separately for **Benchmark 1, Benchmark 2, Benchmark 3, Benchmark 4 and Benchmark 5**, along with the overall average.
- Report **Balanced Accuracy, Precision, Recall/Sensitivity, Specificity, F1-score and Confusion Matrix**.
- Perform at least one additional experiment/ablation study relevant to the assigned project title.
- Clearly document all preprocessing, data splitting, training and evaluation settings.

Important: The five benchmarks should **not be merged into a single dataset for the primary experiments**, unless explicitly required by the assigned project statement.

Each group must independently conduct its experiments. Students shall not use trained models, extracted features, experimental results or code/results generated by another group.

6. Clearly Defined Group-Wise Research Problem Statements

Group Allocation

Sl. No.	Group No.	Reg. No.	Student Name	Title of the Project Group
1	G1	23BVD1004	PILLUTLA PASYANTH	Multi-Representation Learning for Lithography Hotspot Detection
2		23BVD1005	HARIHARAN NAGARAJAN	
3		23BVD1006	MANDAVA RAMYA SREE	
4	G2	23BVD1007	PALEMPALLI LAKSHMI NARASIMHA	DCT-Enhanced Lithography Hotspot Detection
5		23BVD1012	A SANJAY	
6		23BVD1013	ANISH ANIL SHENDE	
7	G3	23BVD1021	UPAMANYU SENGUPTA	Genetic Algorithm-Based Feature Selection for Hotspot Detection
8		23BVD1023	SHARVIL MAHENDRA PINGALE	
9		23BVD1028	PRATYUSH SREEKUMAR	
10	G4	23BVD1030	R SANDHYAA	Encoding-Aware Deep Learning for Lithography Hotspot Detection
11		23BVD1031	SHASHANK CHOWDARY THUMMALA	
12		23BVD1034	S BHARGHAV	
13	G5	23BVD1035	HARI KRISSHNA R	CNN Deep Features with Classical

14		23BVD1036	YOKESH G	Machine Learning Classifiers
15		23BVD1038	TINU P MATHEW	
16	G6	23BVD1039	SUBHAM PRASAD GUPTA	Heterogeneous ML-DL Ensemble for Reliable Hotspot Detection
17		23BVD1042	AKHIL KUMAR KUDIPUDI	
18		23BVD1044	TEJASVINI R	
19	G7	23BVD1045	ADISH VIPIN	Two-Stage Lithography Hotspot Detection Framework
20		23BVD1046	MULAKALA YAGNESH	
21		23BVD1051	AVANEESH KARTHIKEYAN	
22	G8	23BVD1053	RENIGUNTLA SAI MRUDUL	Genetic Algorithm-Optimized CNN for Hotspot Detection
23		23BVD1056	ABHINAV CHAVVA	
24		23BVD1057	SHAIK RAFIQUE AHAMED	
25	G9	23BVD1059	MADISETTY VENKATA LALITHADITYA	Cost-Sensitive Learning for Lithography Hotspot Detection
26		23BVD1060	TANMAY SHARMA	
27		23BVD1061	JAGARAKSHITHA V	
28	G10	23BVD1062	VINAYAK SHREENIVAS SALUNKE	Focal and Hard-Example Learning for Hotspot Detection
29		23BVD1064	HARSH VARDHAN SINGH	
30		23BVD1065	ALAVALA VISHNU KOUSHIK REDDY	
31	G11	23BVD1066	PRIYANUJ BORUAH	Data Balancing and Augmentation Strategies for Hotspot Detection
32		23BVD1067	KARTIK AGGARWAL	
33		24BVD1006	SONAKSHI AGRAWAL	
34	G12	24BVD1023	PALASH ASHUTOSH DURAGKAR	Uncertainty-Aware Lithography Hotspot Detection
35		24BVD1024	VARUN U	
36		24BVD1026	SUBBURAYALU M	
37	G13	24BVD1033	SANJANA CHEJETI	Multi-Objective Genetic Algorithm for Lightweight CNN Design
38		24BVD1036	ADIDEV S	

39		24BVD1041	MISHA MARIA TOMY	
40	G14	24BVD1043	PRANNAVRAJ V S	Accuracy–Latency Optimization of Lithography Hotspot Detectors Accuracy–Latency Optimization of
41		24BVD1048	MAGESH	
42		24BVD1054	VARUN ADITHYA M	
43	G15	24BVD1060	DINESH KUMAR M	Cross-Benchmark Generalization of Lithography Hotspot Detectors
44		24BVD1068	MADHUMITHA M	
45		24BVD1069	KATAKAM SHRIHITA	
46	G16	24BVD1076	EDUPUGANTI VYSHNAVI	Limited-Data Benchmark Adaptation for Hotspot Detection
47		24BVD1084	ABHINAV M A	
48		24BVD1085	MANDA KRISHNA HRIDAY	
49	G17	24BVD1086	KAMALESH S	Robust Lithography Hotspot Detection under Layout Perturbations
50		24BVD1094	HARI GANESH B	
51		24BVD1095	RUPASWATHIKAA S P	
52	G18	24BVD1100	ALLEN JOE A	Explainable AI for Lithography Hotspot Detection
53		24BVD1102	R NAGA ARJUN	
54		24BVD1103	PRASIDDH KUMAR SAHA	
55		24BVD1104	CHITLURI SOHAN	
56	G19	24BVD1106	BONTHALA SAI KIRAN	ML-Based Pre-Screening and Verification Referral for Hotspots
57		24BVD1114	BANDA SIRICHANDANA	
58		24BVD1118	KAMAL SHARMA	
59		24BVD1119	KARTHIKEYAN R V	

Group 1 – Multi-Representation Learning

Research Question: Can combining multiple layout representations improve lithography hotspot detection compared with using a single representation?

Research Focus / Novelty: Investigate complementary feature-extraction/encoding representations and develop a feature/representation fusion strategy.

Minimum Expected Investigation: Compare individual representations against fused representations using the same classification framework. Include an ablation study to identify which representation contributes most.

Group 2 – DCT-Enhanced Hotspot Detection

Research Question: Can frequency-domain information obtained using Discrete Cosine Transform (DCT) improve hotspot detection compared with conventional spatial-domain representations?

Research Focus / Novelty: Investigate spatial features, DCT-based features, and combined spatial-frequency representations.

Minimum Expected Investigation: Quantitatively compare the three settings and determine whether frequency-domain information captures discriminative layout characteristics.

Group 3 – Genetic Algorithm-Based Feature Selection

Research Question: Can Genetic Algorithm (GA)-based feature selection reduce feature dimensionality while maintaining or improving hotspot detection performance?

Research Focus / Novelty: Use GA to select an informative subset of extracted features before classification.

Minimum Expected Investigation: Compare all features, GA-selected features, and a suitable non-optimized/random feature subset. Report dimensionality reduction and performance trade-offs.

Group 4 – Encoding-Aware Deep Learning

Research Question: How does the choice of layout encoding affect CNN-based lithography hotspot detection?

Research Focus / Novelty: Evaluate multiple layout encoding techniques while keeping the CNN architecture and evaluation protocol controlled.

Minimum Expected Investigation: Identify the most effective encoding and explain why the representation influences classification performance.

Group 5 – CNN Deep Features with Classical Machine Learning

Research Question: Can features learned by a CNN and subsequently classified using classical machine-learning models outperform a direct end-to-end classification strategy?

Research Focus / Novelty: Use a CNN as a feature extractor and investigate classifiers such as SVM, Random Forest, Decision Tree, or Logistic Regression.

Minimum Expected Investigation: Compare CNN-feature/classifier combinations against the common baseline and determine whether learned representations transfer effectively to classical classifiers.

Group 6 – Heterogeneous ML-DL Ensemble

Research Question: Can combining predictions from heterogeneous ML/DL models improve hotspot detection reliability compared with an individual model?

Research Focus / Novelty: Develop an ensemble involving complementary models such as CNN, SVM, Random Forest, or ANN using voting, weighted voting, or stacking.

Minimum Expected Investigation: Compare individual models and the ensemble, and analyze which error patterns are reduced by model combination.

Group 7 – Two-Stage Hotspot Detector

Research Question: Can a two-stage detection framework improve hotspot detection by treating difficult or ambiguous samples separately?

Research Focus / Novelty: Develop a fast first-stage detector and a second-stage classifier for samples identified as difficult or uncertain.

Minimum Expected Investigation: Compare single-stage and two-stage systems in terms of balanced accuracy, false negatives, false positives, and computational cost.

Group 8 – Genetic Algorithm-Optimized CNN

Research Question: Can Genetic Algorithm optimization improve the architecture or hyperparameters of a CNN for lithography hotspot detection?

Research Focus / Novelty: Use GA to optimize selected CNN design parameters such as filters, kernel size, depth, learning rate, dropout, or batch size.

Minimum Expected Investigation: Compare the optimized CNN with a manually configured baseline and demonstrate which parameters have the greatest effect.

Group 9 – Cost-Sensitive Hotspot Detection

Research Question: Can explicitly assigning a higher penalty to false-negative hotspot predictions improve practical hotspot detection?

Research Focus / Novelty: Develop a cost-sensitive learning strategy in which missing a true hotspot is penalized more strongly than incorrectly flagging a non-hotspot.

Minimum Expected Investigation: Compare standard and cost-sensitive learning and analyze the trade-off among recall, specificity, precision, F1-score, and balanced accuracy.

Group 10 – Focal / Hard-Example Learning

Research Question: Can a learning strategy that emphasizes difficult and minority samples improve lithography hotspot detection?

Research Focus / Novelty: Investigate focal loss and/or hard-example mining to focus learning on difficult hotspot and misclassified patterns.

Minimum Expected Investigation: Compare conventional training against the proposed strategy and analyze performance specifically on difficult/minority samples.

Group 11 – Data Balancing and Augmentation

Research Question: Which data-balancing strategy provides the most effective improvement for highly imbalanced ICCAD-12 hotspot detection?

Research Focus / Novelty: Systematically compare strategies such as undersampling, oversampling, augmentation, balanced mini-batches, or suitable combinations.

Minimum Expected Investigation: Evaluate not only overall balanced accuracy but also hotspot recall, specificity, F1-score, and the effect of the altered class distribution.

Group 12 – Uncertainty-Aware Hotspot Detection

Research Question: Can model confidence or uncertainty be used to identify patterns that require further verification rather than forcing every sample into a binary decision?

Research Focus / Novelty: Develop a confidence/uncertainty-aware decision mechanism producing high-confidence HS, high-confidence NHS, and uncertain/referral cases.

Minimum Expected Investigation: Study confidence thresholds and quantify the relationship between coverage, error rate, false negatives, and referral rate.

Group 13 – Multi-Objective GA for Lightweight CNN

Research Question: Can Genetic Algorithm simultaneously optimize hotspot detection performance and CNN computational complexity?

Research Focus / Novelty: Formulate a multi-objective optimization problem that maximizes balanced accuracy while minimizing parameters, inference time, or another complexity measure.

Minimum Expected Investigation: Generate and analyze a Pareto front and identify practical accuracy-complexity operating points.

Group 14 – Accuracy-Latency Optimization

Research Question: What is the most suitable trade-off between hotspot detection performance and computational latency?

Research Focus / Novelty: Compare multiple ML/DL models using both predictive performance and computational measurements.

Minimum Expected Investigation: Report balanced accuracy, recall, specificity, F1-score, parameter count and inference time, and identify a practical operating point for CAD-oriented use.

Group 15 – Cross-Benchmark Generalization

Research Question: Does a hotspot detector trained on one ICCAD-12 benchmark generalize effectively to other benchmarks?

Research Focus / Novelty: Perform cross-benchmark experiments in which models trained on one benchmark are tested on other benchmarks.

Minimum Expected Investigation: Construct a 5×5 train/test generalization matrix and identify which benchmark pairs exhibit strong or weak transferability. Analyze possible reasons.

Group 16 – Limited-Data Benchmark Adaptation

Research Question: Can a hotspot detector trained on one benchmark be adapted to a new benchmark using only a small amount of target-domain data?

Research Focus / Novelty: Study source-to-target adaptation using limited target data, such as 5%, 10%, or 20% of the available target training data.

Minimum Expected Investigation: Compare training from scratch with the proposed adaptation strategy and quantify the amount of target data required to achieve useful performance.

Group 17 – Robustness to Layout Perturbations

Research Question: How robust is a hotspot detector when the input layout representation is subjected to controlled perturbations?

Research Focus / Novelty: Create controlled perturbations appropriate to the representation, such as noise, translation, scaling, geometric or encoding variations.

Minimum Expected Investigation: Measure performance degradation from clean to perturbed data and identify which perturbations most strongly affect hotspot detection.

Group 18 – Explainable Hotspot Detection

Research Question: Can explainable-AI techniques identify the layout regions or features that contribute most strongly to hotspot classification?

Research Focus / Novelty: Apply methods such as Grad-CAM, occlusion analysis, saliency analysis, or feature importance to explain hotspot predictions.

Minimum Expected Investigation: Provide visual explanations for representative HS/NHS cases and assess whether the highlighted regions are technically meaningful.

Group 19 – ML-Based Pre-Screening and Verification Referral

Research Question: Can ML-based hotspot detection serve as a fast-pre-screening stage to identify layout patterns that require detailed verification?

Research Focus / Novelty: Develop a decision pipeline in which high-confidence NHS patterns are screened quickly, hotspot/uncertain patterns are flagged, and uncertain cases are referred for further verification.

Minimum Expected Investigation: Evaluate detection performance, referral rate, false-negative rate, and potential screening efficiency. Do not claim actual lithography-simulation speed-up unless simulation evidence is available.

7. Minimum Technical Expectations

- Study and cite at least 5–8 relevant research papers.
- Clearly state the research gap and formulate a testable research question or hypothesis.
- Analyze the ICCAD-12 data distribution and explain the class-imbalance issue.
- Implement the assigned proposed variation independently.
- Perform a controlled baseline-versus-proposed comparison.
- Include at least one ablation study or controlled comparative experiment.
- Report confusion matrix, balanced accuracy, precision, recall/sensitivity, specificity, and F1-score.
- Include appropriate graphs/tables and explain the observed trends rather than merely reporting numbers.
- Where relevant, report inference time, number of parameters, feature dimension, memory, robustness, or cross-benchmark performance.
- Discuss limitations and identify meaningful future research directions.
- Submit reproducible code with a clear README and documented experimental settings.

8. Fifteen-Day Timeline

Day	Activity	Expected Deliverable
Day 1 (05/09/2026)	Understand ICCAD-12, hotspot detection, and the assigned research question	Problem definition
Day 2	Literature survey	Minimum 5 relevant papers identified
Day 3	Dataset and class-distribution analysis	Benchmark statistics and imbalance analysis
Day 4	Baseline implementation/setup	Baseline model running
Day 5	Baseline evaluation	Confusion matrix and baseline metrics
Day 6	Proposed methodology design	Architecture/flowchart and experimental plan
Day 7	Proposed implementation	First working proposed model
Day 8	Model/hyperparameter tuning	Improved proposed model
Day 9	Main experiments	Primary experimental results
Day 10	Ablation/comparative experiments	Comparison table
Day 11	Additional research experiments	Generalization/robustness/complexity/etc., as applicable
Day 12	Quantitative analysis	Graphs, tables and interpretation
Day 13	Research discussion	Findings, limitations and future work
Day 14	Final report and presentation preparation	Complete report and slides
Day 15 (20/09/2026)	Submission	Final report + code + slides

9. Assessment Rubric – 30 Marks

Assessment Component	Marks	Evaluation Criteria
Problem Understanding & Literature Review	5	Understanding of hotspot problem (1); ICCAD-12 understanding (1); relevant literature (1); clearly identified research gap (1). Work allocation table (1);
Baseline Implementation & Experimental Setup	4	Correct preprocessing/data handling (1); baseline implementation (1); appropriate train/validation/test methodology (1); correct evaluation metrics (1).
Proposed Research Method / Novel Variation	6	Clear research question (1); technical correctness (2); meaningful/novel variation (2); justification (1).
Experimental Design & Ablation	6	Baseline vs proposed comparison (2); controlled ablation (1); multiple experimental conditions (2); fair comparison (1).
Results, Metrics & Technical Analysis	6	Confusion matrix and metrics (1); appropriate primary/secondary metrics (1); graphs/tables (1); quantitative comparison (1); technical interpretation – writing (2).
Research Discussion	1	meaning of results; limitations and future work (1).
On-time submission through Microsoft Forms	2	Proper Formatting (1); Submission before deadline (1)
TOTAL	**30**	—

10. Expected Final Submission

- Research report in a conference-paper style format (IEEE).
- Presentation slides.
- Source code / notebook with README and execution instructions.
- Dataset/preprocessing description and experimental configuration.
- Baseline and proposed-model results.
- Confusion matrices, performance tables and relevant plots.
- Ablation/comparative study.
- Clear statement of research gap, contribution, limitations and future work.

11. Important Academic Integrity Requirement

Students may use published papers and publicly available software libraries as references, but must clearly cite all sources. Direct reproduction of an existing paper without a meaningful research variation will not be considered a strong research contribution. Results must not be fabricated, selectively reported, or altered to improve apparent performance.